

Learn-to-learn on Arbitrary Textual Conditioning: A Hypernetwork-Driven Meta-Gated LLM

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Abstract

Conventional LLMs may suffer from corpus heterogeneity and subtle condition changes. While finetuning can create the catastrophe forgetting issue, application of meta-learning on LLMs is also limited due to its complexity and scalability. In this paper, we activate the meta-signal of β within the SwiGLU blocks, resulting in a meta-gating mechanism that adaptively adjusts the nonlinearity of FFN. A hypernetwork is employed which dynamically produces β on textual conditions, providing meta-controllability on LLMs. By testing on different condition types such as task, domain, persona, and style, our method outperforms finetuning and meta-learning baselines, and can generalize reasonably on unseen tasks, condition types, or instructions. Our code can be found in <https://github.com/AaronJi/MeGan>.

1. Introduction

The vast diversity of tasks, styles, domains, and other conditions present in the environment for both human and AI, which raises the problem of meta-learning, or in other words, the capability of ‘learn-to-learn’ given different contexts. While Large Language Models (LLMs) have demonstrated astonishing capabilities across diverse natural language processing (NLP) tasks, they remain relatively deficient in the seamless, dynamic adaptation to those conditions (Reinhart et al., 2025; Zamaraeva et al., 2025). When finetuning on a specific downstream task, their general abilities are often degraded, sometimes termed ‘catastrophic forgetting’ or ‘alignment tax’, which hurts their adaptation capability to new conditions. This performance erosion presents a fundamental challenge to developing robust and versatile AI systems, preventing the application of personalized systems

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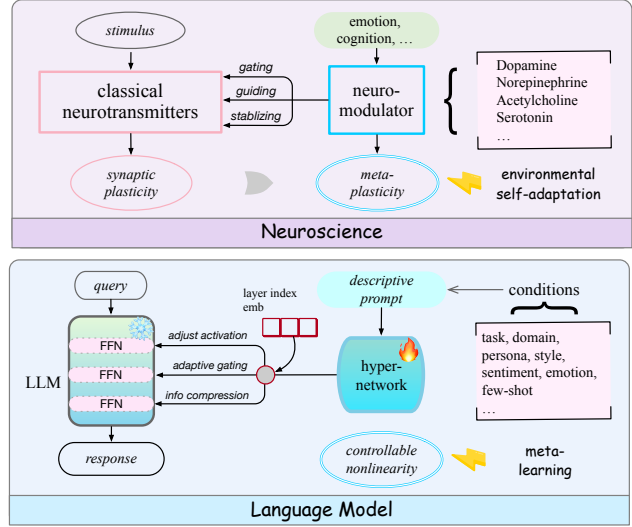


Figure 1. Paradigm of MeGan analogous to neuro-system. Neuro-modulator meta-controls classical neurotransmitters, shaping the synaptic plasticity to meta-plasticity. Similarly, we implement a hypernetwork that converts textual condition inputs into meta signals. These signals are combined with layer index embedding, then meta-control the gating of LLM FFNs.

or expert-level emotional supporters (Srinivas et al., 2025).

Varied meta-training methods are proposed to alleviate these issues, including in-context learning (ICL) (Min et al., 2022; Sinha et al., 2024; Coda-Forno et al., 2023), gradient-based meta-learning (Finn et al., 2017; Song et al., 2020), parameter-efficient fine-tuning (PEFT) (Hu et al., 2022; Tian et al., 2024; Li et al., 2025), and adapter methods (Houlsby et al., 2019; He et al., 2022; Hu et al., 2023). However, ICL usually lags behind finetuning, produces high variance results, resulting in the issue of ‘lost in needle’ by creating a long context window (Ponce & Etchegoyhen, 2025). Gradient or adapter-based methods have high computational overhead in NLP tasks or are difficult to scale up. PEFT methods like LoRA are computationally efficient and have good adaptivity. However, they generally require downstream task-specific training which limits their applications.

In contrast, human nervous system overcomes the aforementioned issues, where the basic stimulus is responded by

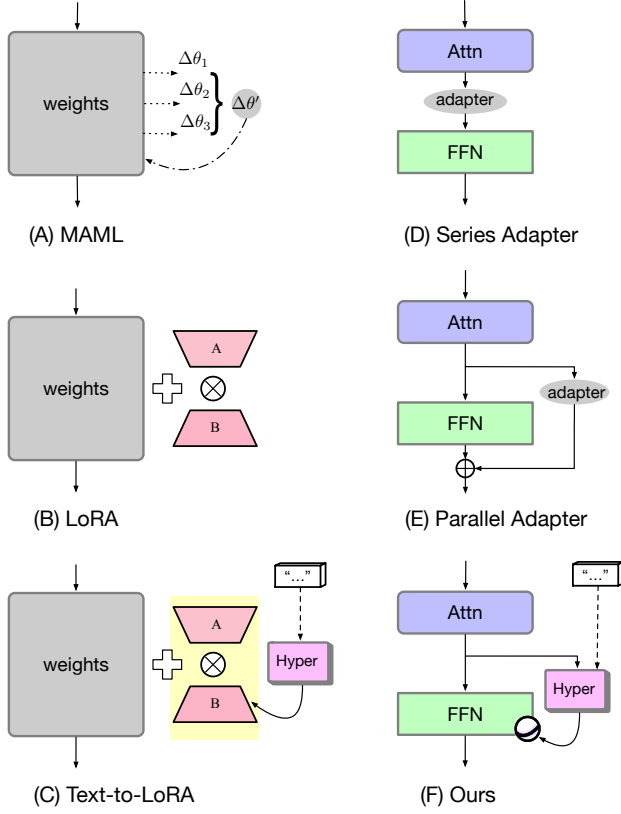


Figure 2. Comparison of meta-learning paradigms on LLMs.

synaptic plasticity based on classical neurotransmitters; and neuromodulator produces the adaptive gain control, through several mechanisms including gating, guidance, and stabilization (Aston-Jones & Cohen, 2005). The system then achieves meta-plasticity, which seamlessly adapts to changing environmental contexts without the metabolic cost of physical rewiring (Servan-Schreiber et al., 1990). Analogously, pretrained LLM can be considered as a generalized query-response processor. Instead of directly prompting or finetuning on it, we implement a hypernetwork with textual *condition* as input, which produces a meta-signal on the original LLM. With original LLM parameters frozen, the hypernetwork is trained by post-hoc adaptation on meta-training samples, which then adapt the LLM conditioned on target tasks (Figure 1).

Similar to Text-to-LoRA (Charakorn et al., 2025), a LoRA-based architecture driven by a text-input hypernetwork (Figure 2 (C)), we also introduce the architecture customization based on textual conditions, to provide higher adaptability. The architecture is similar to the parallel adapter (Figure 2 (E)), by implementing the hypernetwork as a parallel module with FFN, with the cross-attention between LLM latent and textual conditions. Different from adapters which directly control the LLM latent (Qiu et al., 2025), our hypernetwork produces the meta-signal to adaptively adjust

the activations of feedforward network (FFN), resulting in different levels of nonlinearity. This design mirrors the mechanism of the neuron’s f-I (frequency-current) curve (Silver, 2010), and is consistent with previous observations that FFNs encode persona or style semantics earlier and better than self-attention (Poonia & Jain, 2025). To achieve this, we replace the Swish_1 activation with Swish_β , resulting a new type of FFN block called β -SwiGLU. The hypernetwork produces β by a bottleneck structure, extracting key conditional patterns by information compression (Schmidhuber, 2009; Sutskever, 2023; Delétang et al., 2024), replacing the original self-gating by meta-gating. Correspondingly, the entire framework is named by **Meta-Gating-on-Condition (MeGan)**.

MeGan is trained by supervised fine-tuning (SFT), with an auxiliary L-2 regularization on MLP weights. We experiment it on different settings of meta-learning, with different meta-conditions such as task, domain, persona, style, and sentiment. Our framework outperforms prompting finetuning, or meta-learning baselines, showing good zero-shot performance on unseen tasks or condition types. Further analysis shows that MeGan is scalable, robust and parameter-efficient. Distribution of β also reveals that MeGan adaptively captures the semantics of conditions by dynamic β calculation. To summarize, our main contributions include:

- We propose MeGan, a meta-gating framework on meta-learning, which mirrors neuroscience mechanism.
- We implement a novel hypernetwork, which converts arbitrary conditions into a self-adaptive meta-signal, steering the nonlinearity of β -SwiGLU blocks.
- Thorough theoretical studies and empirical experiments are conducted to verify our methodology’s performance, stability, and scalability.

2. Related Work

While transformer has proven to be a robust general-purpose architecture and also a few-shot learner, it still struggles in complicated, unseen tasks. Meta-learning, which deals with such types of problems, is therefore investigated on LLMs. Below, we discuss several categories of paradigms in comparison to our method (Figure 2).

Adapter methods. To avoid the overfit issue of finetuning, studies are conducted by implementing an extra adapter alongside the main LLM backbone, with the main LLM parameters unchanged. Depending on the locations, they can be classified into series adapter (Houlsby et al., 2019; Qiu et al., 2025) and parallel adapter (He et al., 2022)(Figure 2 (D) and (E)). LLM-Adapters (Hu et al., 2023) systematically experiment with different types of adapters, and

Table 1. Methodology summaries of the baselines and MeGan. *: dynamic low-rank reparameterization, which is similar to LoRA.

Method	Structure	Meta tasks		Target tasks		Original param kept
	customization	adaptation	update	adaptation	adjustment	
LoRA (Hu et al., 2022)	✗	✗	LoRA	zero-shot	✗	✓
SFT	✗	✗	FT	zero-shot	✗	✗
CMAML (Song et al., 2020)	pruning	gradient	meta-update	zero-shot	✗	✗
Meta-in-context						
meta-icl (Coda-Forno et al., 2023)	✗	in-context	✗	few-shot	✗	✓
MetaICL (Min et al., 2022)	✗	in-context	FT	few-shot	✗	✗
MAML-en-LLM (Sinha et al., 2024)	✗	in-context	meta-update	few-shot	✗	✗
Meta-on-LoRA						
MLtD (Hou et al., 2022b)	struc. control	PEFT*	meta-update	zero-shot	PEFT*	✗
HydraLoRA (Tian et al., 2024)	✗	asym. LoRA	MoE routing	merged experts	LoRA	✓
Meta-LoRA (Li et al., 2025)	✗	grad. similarity	reweighting	zero-shot	LoRA	✓
Text-to-LoRA (Charakorn et al., 2025)	hypernetwork	task-wise text	FT / Recon	task-wise text	LoRA	✓
Meta-on-Gating (Ours)						
MeGan	hypernetwork	sample-wise text	FT	sample-wise text	activation	✓

conclude that the parallel adapter can adapt well to versatile and changing tasks.

Parameter-efficient fine-tuning (PEFT) methods can also be considered as a special type of architecture-agnostic adapters, such as LoRA (Hu et al., 2022) (Figure 2 (B)). However, LoRA requires a specific downstream training set, which may not be viable in the meta-learning case. Studies also show that the performance LoRA may suffer from heterogeneity corpus (Tian et al., 2024).

Meta-in-context methods. Direct application of traditional meta-learning such as MAML (Finn et al., 2017; Song et al., 2020) on NLP tasks is limited, due to its complicated bi-level optimization (Figure 2 (A)). Instead, many meta-learning studies on LLMs incorporate the in-context information, including MetaICL (Min et al., 2022), MetaICT (Chen et al., 2022), meta-in-context learning (Coda-Forno et al., 2023), MAML-en-LLM (Sinha et al., 2024) and MICRE (Li et al., 2024). In this work, we show that ICL can fail for some specific adaptation conditions.

Meta-on-LoRA methods. Another category leverages the adaptation of low-rank parameters of LoRA. Such methods include MLtD (Hou et al., 2022b), which applies MAML on LLM with either task-adaptive reparameterization (TARP) or task-adaptive model structures (TAMS), hydraLoRA (Tian et al., 2024), which uses asymmetric LoRA to route MoE, and also Meta-LoRA (Li et al., 2025), which conducts sample reweighting by gradient similarity.

Meta with customized structures. There is also a special work called Text-to-LoRA (Charakorn et al., 2025) (Figure 2 (C)), which implements a hypernetwork, that receives textual task descriptions and outputs the LoRA parameters. Its hypernetwork can be trained by the FT loss or a reconstruction loss with pretrained LoRA. In this paper, we propose

another meta-control paradigm for LLMs, which leverages a novel gating mechanism, the meta-gating (Lin et al., 2021; Hou et al., 2022a; 2023) (Figure 2 (F)). Replacing the original self-gating in the FFN, we let our gating to be adaptive with a hypernetwork on auxiliary conditions, similar to Text-to-LoRA. We employ the FT loss since (Charakorn et al., 2025) observes that the reconstruction loss fails to generalize on unseen tasks. We also have training-inference-consistent adaptation on text instructions.

Table 1 summarizes all the aforementioned methods and highlights the difference of our methodology.

3. Preliminary

3.1. LLM Modules

Activation. Starting from a generalized β -Sigmoid,

$$\sigma_{\beta}(x) := 1/(1 + \exp(-\beta x)) \quad (1)$$

the famous Sigmoid $\sigma(x)$ is its special case ($\beta = 1$). Based on Sigmoid, the Swish $_{\beta}$ ($\beta > 0$) and its special form SiLU ($\beta = 1$) has the following formulation:

$$\text{Swish}_{\beta}(x) = x * \sigma_{\beta}(x) \quad (2)$$

$$\text{SiLU}(x) := \text{Swish}_1(x) = x * \sigma(x) \quad (3)$$

As β increases, the nonlinearity of Swish $_{\beta}$ is amplified. At two extreme situations, Swish $_{\beta}$ becomes Linear when $\beta \rightarrow 0$, and ReLU when $\beta \rightarrow +\infty$. Modern LLMs (LlaMA, Mistral, Qwen, etc) usually adopt SiLU as a standardized implementation to reduce pretraining complexity.

Self-Gating. Within the feedforward network (FFN) blocks of LLM, GLU (Gated Linear Units) structure is adopted which uses a gating network to reweight the MLP output. By using Swish $_1$ (SiLU) as the activation function

Table 2. Summaries of the meta-conditions z , including task, domain, persona, style, sentiment, etc. We provide a generalized framework that self-adapts to all the conditions above. Types of tasks are marked in *Italic*. More prompts are in Appendix A.1.

Type	Attribute	Prompt	Related Datasets
task	<i>summarization/QA/classification/...</i>	(depending on detailed task; see Appendix A.4)	CrossFit&UnifiedQA , SNI
domain	debate/science/email, ...	Please provide the summarization on the domain of {domain}.	AdaptSum (<i>summarization</i>)
persona	["i like to remodel homes." " my favorite holiday is halloween."]	Please provide the response with the knowledge of {persona}.	Persona-Chat (<i>dialogue</i>)
style	formal/informal moral/immoral	Please provide the response with the style of {style}.	GYAFC (<i>dialogue</i>) MIC (<i>QA</i>)
sentiment	positive/neutral/negative	Please provide the response with the sentiment of {sentiment}.	SST, Amazon, IMDB (<i>review</i>)
emotion	joy/sad/anger/...	Please provide the response with the emotion of {emotion}.	EmoryNLP, DailyDialog (<i>dialogue</i>)

of gating, the FFN block has the **SwiGLU** architecture:

$$y = W_{\text{down}} (\text{Swish}_1(W_{\text{gate}}x) \otimes (W_{\text{up}}x)) \quad (4)$$

$$W_{\text{up}}, W_{\text{gate}} \in \mathcal{R}^{D \times C}, W_{\text{down}} \in \mathcal{R}^{C \times D}, C > D$$

where $\otimes$ denotes the element-wise multiplication, D is the hidden size, and C is the FFN intermediate size. In practice, C is usually a multiple of D . SwiGLU can also be considered as **self-gating** which means the same latent is both inputs of the MLP and the gating network.

3.2. LLM architecture

LLM is a decoder-only transformer with interleaved self-attention and FFN layers:

$$e_0 = \text{emb}(x) \in \mathcal{R}^D, \text{logit} = \text{proj}(e_L)$$

$$e_l = \text{FFN}(\text{self-attn}(e_{l-1})), l = 1, \dots, L \quad (5)$$

where x is the text input, L is the total number of layers, e_l is the latent on the l -th layer, and *logit* is the final output logit. $\text{emb}(x)$ represents the tokenization and embedding layer, while proj is the final projection layer.

3.3. Hypernetworks

Given a base model $y = G_w(x)$, a hypernetwork (Ha et al., 2017) can be formed with w generated by another auto-adaptive network: $w = \mathcal{H}_\theta(z)$ where z is the adaptive input, and θ is the hypernetwork trainable parameter.

4. Problem Formulation

4.1. Data format

The problem of meta-training can be formulated on a collection of tasks $\mathcal{T}_{1:C}$, where C is the total number of meta-tasks. For each task, the data has a format (x, y) , where x denotes the input, y denotes the output. During the training, the

model learns to condition on the samples of a subset of tasks

$$\mathcal{T}_i = \{(x_j^i, y_j^i)\}_{j=1}^{N_i}, \forall i \in [1, C] \quad (6)$$

where N_i is the sample number of the i -th task. During the test stage, the model needs to adapt to an arbitrary task $\mathcal{T}^*$

$$\arg \max P(y^*|x^*), (x^*, y^*) \in \mathcal{T}^* \quad (7)$$

with two typical settings of target tasks in meta-learning:

- i) Low resource (LR): $\mathcal{T}^*$ belongs to the C meta-training tasks, but with relatively limited training samples.
- ii) Unseen (US): $\mathcal{T}^* \notin \mathcal{T}_{1:C}$, outside the scope of LLM.

In Section 6, we investigate both settings to provide a comprehensive understanding of model meta-performances.

4.2. Meta-on-conditions

Compared to baselines in Table 2, in this paper, we propose that meta-learning can also be conducted on **generalized textual conditions**. Such conditions can be readily obtained, especially in the meta-learning setting, where the model adaptation is frequently required on the switch of different types of meta-conditions, such as task, domain, persona, style, sentiment, and emotion. These conditions, however, are insufficiently studied by previous meta-learning methods except Text-to-LoRA¹.

To learn-to-learn on such conditions, we reformulate such datasets with the format of (x, y, z) , where z is the auxiliary annotation of those conditions. During the meta-training, z is first inserted into different instructional prompts (depending on the condition type), then is used to adapt the model. During inference, the model produces the best output conditioned on the test z : $\arg \max P(y^*|x^*, z^*)$. In this

¹However, their experiment solely explores the adaptation on task-wise description.

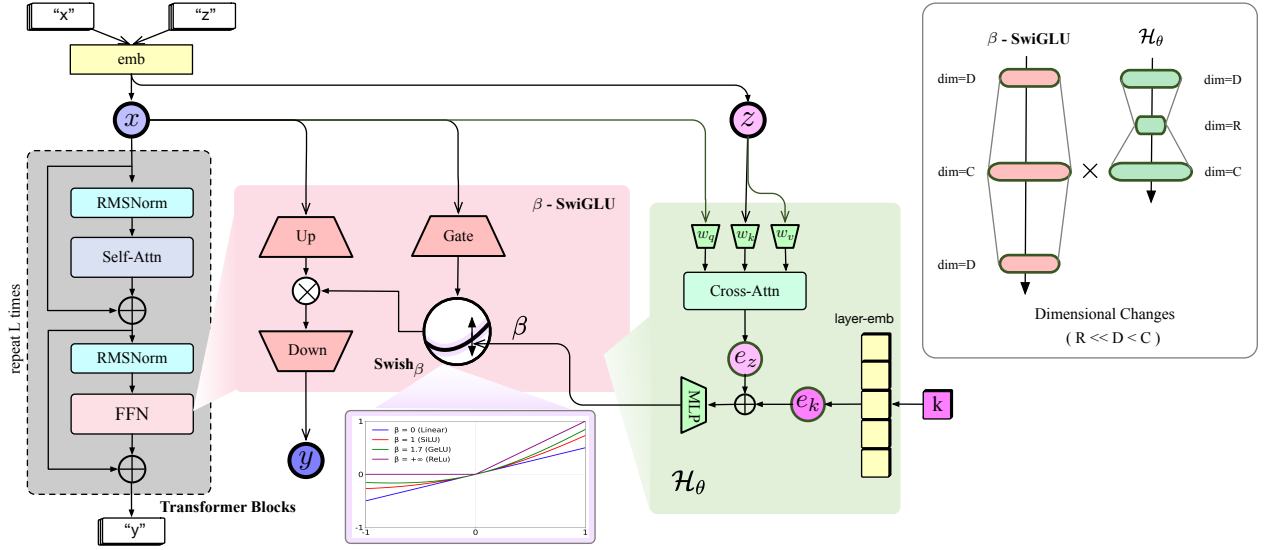


Figure 3. Framework of MeGan. (x, y, z) indicate the input, output, and condition, respectively. $k \in [1, K]$ denotes the layer index.

formulation, z can semantically bridge the meta information between training and target tasks. Table 2 summarizes different condition types, possible attributes, example prompts, and typical relative datasets.

5. Method

For problem proposed in Section 4.2, we introduce a novel meta-gating architecture, which is initialized from the original self-gating, driven by a hypernetwork with condition inputs. Post-hoc adaptation is then conducted on meta-tasks.

5.1. Meta-Gating

Beta-SwiGLU. To provide an extra DoF (degree of freedom) for meta-control, we reactivate the β of SiLU ($\text{Swish}_{\beta=1}$ function (Eq (3)); accordingly, the SwiGLU (Eq (4)) becomes β -SwiGLU, with Swish_1 replaced by Swish_{β} :

$$y = W_{\text{down}} (\text{Swish}_{\beta} (W_{\text{gate}} x) \otimes (W_{\text{up}} x)), \beta > 0 \quad (8)$$

where β dynamically adjusts the slope of the Swish function.

Apparently, when $\beta = 1$, β -SwiGLU decays to the original SwiGLU. Also, the gradient of Swish_{β} is bounded by $1/4|x|^2$ with detailed derivation in Appendix A.2. This property ensures that training β -SwiGLU is friendly.

The hypernetwork. We implement a lightweight hypernetwork which projects the textual condition z into β on each layer, denoting $\mathcal{H}_{\theta}(z, l)$. We first replace the placeholder of prompt p (examples in Table 2) with z , then process it with the tokenizer and embedding layer of the LLM itself. $\mathcal{H}_{\theta}(z)$ then implement a cross attention with a reduced dimension R , added with layer embedding, and finally feed into a one-layer, bias-free, MLP. In more details,

on the l -th layer, the corresponding β^l is generated by

$$z = \text{emb}(p(z)) \quad (9)$$

$$e^z = \text{cross-attn}(q=W^q z, k=W^k e^l, v=W^v e^l)$$

$$\beta^l = W(e^z + \text{layer-emb}(l)), l = 1, 2, \dots, L$$

$$W^q, W^k, W^v \in \mathcal{R}^{D \times R}, W \in \mathcal{R}^{R \times C}$$

where MLP is one-layer, bias-free, has an input-output dimension of (R, C) , and includes a final activation of $\tanh$.

5.2. The Entire Model

Re-formularized LLM. We replace the SwiGLU of LLM with our β -SwiGLU(x), with β dynamically determined by the hypernetwork. The model can then be expressed as

$$y = \mathcal{G}_{\{w, \beta^l = \mathcal{H}_{\theta}(z, l)\}}(x) \quad (10)$$

where θ is the union of trainable parameters of cross-attn and MLP of $\mathcal{H}$. By such a formulation, we disentangle the meta-condition adaptation from the foundation capability, by learning of β and w , respectively. Figure 3 visualizes the entire framework.

Number of parameters. MeGan has a relatively small set of trainable parameters, which ensures its computational efficiency. By utilizing the embedding layer of the LLM itself, we avoid using an extra embedding model (which is done in Text-to-LoRA), reducing the parameters and helping the information be more aligned. # param of MeGan has a **linear** dependency on the reduced hidden size R , which is less than Text-to-LoRA, according to their reported # param. Appendix A.2 shows the detailed derivation.

Table 3. Statistics of meta-training datasets used in this paper. ‘Cond.’ denotes condition. ‘LR’ and ‘US’ indicate low resource and unseen. ‘per.’ denotes persona and ‘senti’ denotes sentiment.

Dataset	Meta-train		Target		
	Cond.	# num	setting	Cond.	# num
Per-Chat	1,137 per.	8.9K	US	100 per.	0.97K
AdaptSum	6 domain	1.5K	LR	6 domain	8.0K
CrossFit-			LR	80 task	186.6K
&UnifiedQA	325 task	3.54M	US	11 task	4.9K
SNI	479 task	1.44M	US	10 task	23.3K
GYAFC			US	2 style	1.6K
MIC			US	2 style	10.9K
SST			US	5 senti.	1.0K

Information compression. By the design of $\mathcal{H}_\theta$, we disentangle the shared meta control and the layerwise difference from a modularity perspective. $\mathcal{H}_\theta$ can also be considered as a **information bottleneck** hypernetwork, with the latent size first compressed by the attention from D to ($R \ll D$), then projected back to C by MLP. This design not only reduces the number of trainable parameters, but also helps compress the information and extract the key patterns (Schmidhuber, 2009; Sutskever, 2023; Delétang et al., 2024). Figure 3 highlights this dimension configuration. Appendix A.2 also provides an informative theory analysis.

5.3. Training Mechanism

We train on samples (x, y, z) , with the original LLM parameters w frozen. Based on the following losses, the gradient of θ is automatically calculated and back-propagated.

The cross-entropy loss. This cross-entropy (CE) loss term is the standard SFT loss on output tokens y , with masks on input tokens x : $\mathcal{L}_\theta^{ce} = \sum yP(\mathcal{G}_{\{w, \beta_\theta(z)\}}|x)$.

Table 4. Results of B-2 and R-L on Persona-Chat and AdaptSum. The best result is marked **bolded** and the second-best is underlined. *: results from original published papers.

Method	Persona-Chat		AdaptSum	
	R-L	B-2	R-L	B-2
CMAML*		1.70		
MLtD*		1.20	37.04	
LLaMA3-8B-Instruct	11.79	3.64	13.35	4.72
+ CoT	7.84	1.91	10.98	4.21
+ PS	9.50	2.66	12.15	4.04
+ MP	8.31	2.04	10.07	3.32
+ ICL	13.75	5.23	23.35	11.29
+ LoRA	21.35	9.63	13.37	4.75
+ SFT	22.97	9.94	20.99	8.80
+ meta-icl	15.74	6.73	13.03	4.37
+ Text-to-LoRA	21.75	13.70	20.02	7.98
+ MeGan (ours)	23.15	10.15	41.85	26.88

Regularization. To ensure β does not deviate from 1 too much, we add an auxiliary loss term to have a 1-2 regularization²: $\mathcal{L}_\theta^{reg} = |\beta_\theta(z)|_2$.

The total loss. The total loss is then a simple linear weighted sum of them

$$\min_{\theta} \mathcal{L} = \mathcal{L}_\theta^{ce} + f \mathcal{L}_\theta^{reg} \quad (11)$$

where f is the regularization weight. Finally, we conduct full finetune (FT) on loss $\mathcal{L}$.

6. Experiments

6.1. Implementation

The model backbone is Llama-3.1-8B-Instruct (Grattafiori et al., 2024), which has $K = 32$ layers, hidden size $D = 4096$, and intermediate size $C = 14336$. We implement $\mathcal{H}$ with reduced hidden size $R = 128$, resulting in 3,411,968 total parameters. We set the regularization weight $f = 0.001$. Other configurations can be found in Appendix A.7.

6.2. Datasets

MeGan is trained by meta-learning on a mixture of datasets, conditioned on different textual information:

- Persona: we use Persona-Chat (Zhang et al., 2018), which conditions the dialogue on unseen persona profiles.
- Domain: AdaptSum (Yu et al., 2021) is employed to conduct the summarization task on 6 domains: debate, dialogue, email, movie review, science, social media, with low-resource training samples.
- Task: we first adopt the experimental setting in (Min et al., 2022; Sinha et al., 2024) which employ the mixture of CROSSFIT (Ye et al., 2021) and UNIFIEDQA (Khashabi et al., 2020), covering a variety of tasks, and is classified into six transfer settings: HR→LR, Class→Class, non-Class→Class, QA→QA, non-QA→QA, non-NLI→NLI, and non-Para→Para; the second experiment setting refers to (Charakorn et al., 2025), which uses the Super-NaturalInstructions (SNI) (Wang et al., 2022a) as training and validation sets, while tests the model on ten generalized mainstream benchmarks, including Arc-challenge (ArcC), Arc-easy (ArcE), BoolQ (BQ), Hellaswag (HS), OpenbookQA (OQA), PIQA, Winogrande (WG), GSM8K, HumanEval (HE), and MBPP.

Detailed data splits and statistics are in Table 3. More dataset and preprocessing details are in Appendix A.4.

²To utilize the regularization directly, we implement Swish_{1+ β} in practice, where $\beta = 0$ corresponds to the raw LLM. See Appendix A.2 for more details.

Table 5. Results on CrossFit&UnifiedQA. Two numbers indicate the average performances on the entire test set and the unseen task test set. HR denotes high resource, Class denotes Classification, NLI denotes natural language inference, and Para denotes paraphrase. *: results from original published papers.

Method	HR →LR	Class →Class	non-Class →Class	QA →QA	non-QA →QA	non-NLI →NLI	non-Para →Para
MetaICL*	43.3/35.3	43.4/32.3	38.1/28.1	46.0/69.9	38.5/48.3	49.0/80.1	33.1/34.0
MAML-en-LLM*	48.0/50.9	51.1/49.0	50.5/46.8	42.5/55.6	40.0/47.1	52.4/65.0	53.3/58.0
Direct	13.7/1.9	5.5/1.9	5.5/1.9	20.1/28.5	20.1/28.5	60.2/56.3	66.6/32.8
LoRA	61.7/60.3	61.1/60.8	59.7/58.8	23.8/23.9	42.0/42.0	62.4/63.5	78.6/77.8
SFT	55.5/54.0	56.4/55.7	59.4/59.2	27.0/27.1	44.5/55.5	65.3/66.4	79.1/78.3
MeGan (ours)	72.2/70.9	64.3/52.6	60.5/49.7	37.8/57.0	72.3/79.5	70.9/81.4	79.1/44.9

Table 6. Zero-shot performance of methods with SNI as the training set. *: results directly obtained from Charakorn et al. (2025).

	ArcC (acc)	ArcE (acc)	BQ (acc)	HS (acc)	OQA (acc)	PIQA (acc)	WG (acc)	GSM8K (acc)	HE (pass@1)	MBPP (pass@1)	Avg.
Direct*	73.3	90.6	80.4	66.6	75.4	79.8	55.3	75.7	66.5	68.7	73.2
ICL*	80.7	91.9	80.0	59.3	77.6	80.9	61.3	75.7	66.5	70.4	74.4
Prepending task desc.*	80.2	92.5	79.9	69.8	78.4	81.7	62.4	75.7	68.3	70.2	75.9
LoRA*	82.0	92.8	83.3	70.8	81.8	83.8	60.3	77.6	63.4	69.4	76.5
SFT	80.9	93.1	79.7	66.3	78.6	81.3	58.2	76.4	65.2	63.3	74.3
Text-to-LoRA*	82.4	92.9	84.4	72.8	81.8	81.2	60.0	79.1	64.6	69.9	76.9
MeGan	85.3	94.2	83.0	77.5	84.2	85.8	67.5	74.1	57.9	65.2	77.5

Table 7. Ablation on Persona-Chat and AdaptSum. p denotes the prompt and w denotes the original LLM parameters.

Method	Persona-Chat			AdaptSum		
	R-L	B-2	D-2	R-L	B-2	D-2
w/o p	20.38	8.98	0.21	0.18	0.04	0.01
w/o pos-emb	7.11	2.22	0.04	16.82	6.93	0.14
w/o $\mathcal{L}_{reg}$	20.38	10.15	0.11	36.62	21.63	0.34
MeGan	23.15	10.15	0.22	41.85	26.88	0.35

6.3. Baselines

MeGan compared to several types of baselines, including (1) Direct inference and Prompt-based methods such as in-context learning (ICL), CoT (Wei et al., 2022), Plan-and-Solve (PS) (Wang et al., 2023), and Metacognitive Prompting (MP) (Wang & Zhao, 2024). These methods do not adapt parameters to specific meta-tasks, therefore naturally avoids the overfit. However, they are expected to have **relatively poor adaptation**.

(2) Finetuning methods such as SFT and LoRA (Hu et al., 2022). They are not designed specifically for meta-learning, therefore expected to adapt worse. LoRA is expected to have **less alignment tax** than SFT by disentangling the original parameters and newly-adapted low-rank matrices.

(3) Meta-in-context methods such as meta-in-context learning³ (Coda-Forno et al., 2023), CMAML (Song et al., 2020), MLtD (Hou et al., 2022b), MetaICL (Min et al., 2022), and

³We name it by meta-icl for shot in the following contexts.

MAML-en-LLM (Sinha et al., 2024), all of which leverage the few-shot examples to conduct meta-learning on LLM. (4) Text-to-LoRA (Charakorn et al., 2025) which implements a hypernetwork that converts task descriptions to low-rank parameters (A, B) in LoRA.

Further introductions and implementation details of baselines can be found in Appendix A.5.

6.4. Metrics

For generation tasks, we consider similarity-based metrics such as famous Rouge-L (**R-L**) (Lin, 2004) and BLEU-2 (**B-2**) (Papineni et al., 2002), as well as Dist-2 (**D-2**) (Li et al., 2015), which indicates the response diversity. For multi-choice, classification, or mathematics tasks, we calculate the answer accuracy (**acc**) compared to the truth options. For coding tasks, we report the **pass@1** results. Detailed definitions are in Appendix A.6.

6.5. Results

MeGan adapts well to low-resource domain or persona. Table 4 shows the results on Persona-Chat and AdaptSum. Prompt-based methods (except ICL) can not adapt to target responses (R-L and B-2) but keep the response diversity (D-2). On the other hand, in-context learning methods (ICL and metaICL), finetuning-based methods (LoRA and SFT) and Text-to-LoRA have improved performances, suggesting the meta capabilities on new personas or domains. Finally, our MeGan achieves the best performance on both bench-

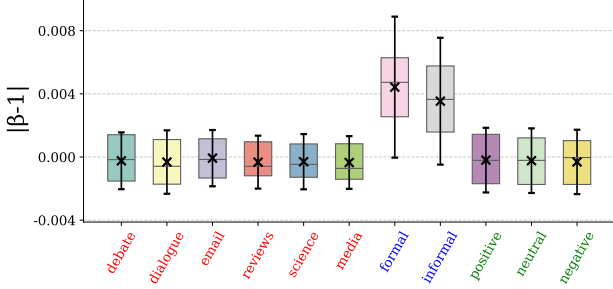


Figure 4. Distribution of β with respect to textual conditions.

marks, indicating that the meta-gating mechanism helps the adaptation to these conditions.

Zero-shot test on styles and sentiments. Appendix B.1 shows detailed zero-shot results on GYAFC, MIC, and SST. Appendix B.2 further exhibit the corresponding good cases of MeGan as well as on typical open-ended cases. All these promising results verify that MeGan can also adapt to new types of textual conditions such as styles and sentiments.

MeGan can generalize on low-resource or unseen tasks.

Table 5 shows the performances on CrossFit&UnifiedQA, which evaluates the adaptation capability from on low-resources or unseen target tasks. MeGan again outperforms all baselines, suggesting that our method also helps the generation conditioned on the task types.

MeGan adapts well on generalized benchmarks. Table 6 exhibits the results on 10 general benchmarks, with models trained by the SNI dataset. MeGan performs the best on ArcC, ArcE, HS, OQA, PIQA, WG, and the averaged score, indicating that our methodology can also work well on different general benchmarks.

Appendix B.5 shows our results based on Mistral-7B-instruct (Jiang et al., 2023), also trained by SNI and evaluated on those benchmarks, which provides an apple-to-apple comparison to the main result of Charakorn et al. (2025).

Ablations. We conduct ablation studies by removing each component of MeGan, including the meta-prompt p , the positional embedding, and the regularization loss $\mathcal{L}_{reg}$. Table 7 shows the ablation results in which the standard MeGan still performs the best, indicating each component is necessary.

7. Discussion

7.1. Distribution of beta

Figure 4 compares the distribution of β on different textual conditions, with $\beta = 1$ aligned to the original LLM. One can find that the FFN is customized by different activation

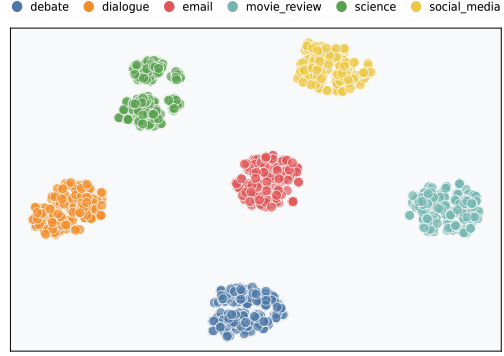


Figure 5. t-SNE analysis of averaged β on AdaptSum.

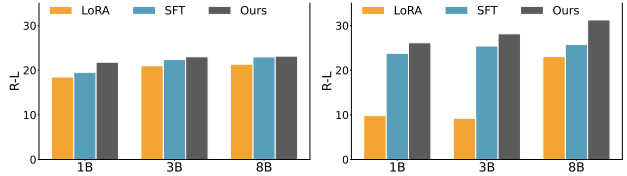


Figure 6. Result comparisons on different model sizes for Persona-Chat (left, low-resource) and GYAFC (right, zero-shot).

slopes, according to different conditions. By checking the sentiment conditions, one can find that the distribution of ‘neutral’ lies between ‘positive’ and ‘negative’, suggesting that this activation customization is also aligned with human commonsense. Figure 5 conducts t-SNE analysis of averaged β on AdaptSum, where evident grouping can be observed w.r.t different domains.

7.2. Scalability

Figure 6 compares MeGan to LoRA and SFT on Persona-Chat and GYAFC, with respect to different model sizes. As model size decreases, MeGan’s performance degrades less than the other two methods, indicating that it is less affected by the foundation model’s capabilities. Note that this phenomenon is more pronounced in the zero-shot case, highlighting the effectiveness of MeGan for meta-learning.

8. Conclusion

In this paper, we propose MeGan, which conducts meta-control on LLMs with FFN gating adaptive to different conditions, such as task, domain, persona, style, and sentiment. We replace the SiLU activation in LLM by Swish_β , resulting in the new FFN block called β -SwiGLU. We further implement a hypernetwork which automatically converts textual conditions into β , resulting a meta-gating mechanism, adjusting the nonlinearity of activations upon different situations. This hypernetwork is then generalized on target tasks with dynamic condition comprehension. MeGan exhibits state-of-the-art performance on meta-learning.

The Impact Statement

This paper presents work whose goal is to advance the field of meta-learning on LLMs. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. More Implementation Details

A.1. Detailed Instructions

Table 8 list more prompts for different datasets.

Table 8. Instructions for varied datasets.

Dataset	Instructions
Persona-Chat	You are engaged in a conversation with the user. The user have the following profiles. You should consider the profiles and make the corresponding appropriate response. profiles: {profile}
	You are chatting with a user, who has the following profiles. Consider the profiles and respond. profiles: {profile}
	Provide the appropriate response based on the user profiles. profiles: {profile}
AdaptSum	You are dealing with an abstractive summarization with different domains. You should consider the following domain tag, and adjust your summarization accordingly. domain: {domain}
	You are dealing with an abstractive summarization with different domains. Adjust your summarization accordingly. domain: {domain}
	Conduct the summarization based on its specific domains. domain: {style}
SST	Please answer the question with the sentiment of {sentiment} .
	Please answer the question with following sentiment. sentiment: {sentiment}
	Reply with sentiment: {sentiment}
GYAFC / MIC	Please answer the question with the style of {style} .
	Please answer the question with following style. style: {style}
	Reply with style: {style}

For CrossFit&UnifiedQA, we use the prompt from PromptScience⁴ as the task descriptions.

A.2. Details of Model

Swish’s design was inspired by the use of sigmoid functions for gating in LSTMs and highway networks.

Gradient of Sigmoid. The gradient of Sigmoid has the following property:

$$\sigma'(x) := \sigma(x)(1 - \sigma(x)) \quad (12)$$

Gradient of β -SwiGLU. Our implementation introduces β -SwiGLU, which revises the LLM architecture, with new back-propagation path. To ensure the training convergence and stability, here we derive the gradient of Swish $_{\beta}$ and indicate the boundedness. Given its expression Swish $_{\beta} = \frac{x}{1+\exp(-\beta x)}$, based on the Sigmoid gradient formula (Eq. 12), we derive its gradient

⁴<https://github.com/bigscience-workshop/promptsources>

$$d(\text{Swish}_\beta)/d\beta = x^2 \sigma_\beta(x)(1 - \sigma_\beta(x))$$

where $\sigma_\beta(x) \in (0, 1)$. Obviously, the gradient norm of Swish_β is bounded by

$$|d(\text{Swish}_\beta)/d\beta| \leq 1/4|x|^2 \quad (13)$$

Detailed implementation of Swish_β . By not losing generality, here we replace β in Equation 2 with $1 + \beta$, for ease of derivation simplicity.

The official code of Pytorch does not provide an explicit manner to input β in Swish. Instead, we apply the following property of Swish

$$\beta \text{Swish}_\beta(x) = \text{Swish}_1(\beta x) \quad (14)$$

and actually implement SwiGLU_β with the official function of SiLU by the following formulation:

$$\text{output} = W_{\text{down}} \left(\frac{\text{Swish}_1((1 + \beta)(W_{\text{gate}} \cdot x))}{1 + \beta} \otimes (W_{\text{up}}x) \right)$$

SwiGLU. The gating mechanism is also employed in LLM, with the same value as the input of Swish as well as the gating network, which is called self-gating. Self-gating is usually believed to have positive derivatives, unboundedness, and smoothness, preventing gradient vanishing and gradient explosion.

Given the famous GLU (Gated Linear Units) structure:

$$\text{GLU}(x, W, V, b, c) = \sigma(Wx + b) \otimes (Vx + c) \quad (15)$$

where $\otimes$ denotes the element-wise multiplication, we then obtain SwiGLU which implements σ in GLU with Swish_1

$$\text{SwiGLU}(x, W, V) = \text{Swish}_1(Wx) \otimes (Vx) \quad (16)$$

where we omit biases b, c for simplicity. Modern LLMs employ SwiGLU to form FFN blocks (AI@Meta, 2024; Qwen Team, 2024), since self-gating has positive derivatives, unboundedness, and smoothness, preventing gradient vanishing and gradient explosion.

Trainable parameter calculation. In more details, the parameter number can be calculated as:

$$\begin{aligned} |\theta|_{\text{cross-attn}} &= 3(D * R) \\ |\theta|_{\text{layer-emb}} &= L * R, |\theta|_{\text{mlp}} = R * C \\ |\theta| &= (3D + L + C) * R \end{aligned}$$

Therefore, # param of MeGan has a **linear** dependency on the reduced hidden size R .

Take LLaMA3-8B as an example, with $L = 32, D = 4096, C = 14336$, then $|\theta| = 26656R$. With R set to 128, our model has $|\theta| = 3,411,968$ parameters, which is smaller than another lightweight hyperparameter baseline, Text-to-LoRA (its small architecture has 4,923,392 parameters) (also based on an 8B LLM backbone).

Discussion of Bottleneck Hypernetwork on Information Theory. Our bottleneck hypernetwork is also aligned with “**Information Bottleneck Hypothesis for Deep Learning**,” (Tishby & Zaslavsky, 2015) with the core proposition summarized as:

The training process of a deep neural network inherently implements the information bottleneck principle in a layered fashion.

Their Argument and Key Observations:

Algorithm 1 Training of MeGan

Input: a pretrained LLM G_w ; a stylized prompt p

Parameter: batch size b , reg-weight f

Output: meta-param θ

- 1: Initialize w from G_w ; detach $w.\text{grad}()$
 - 2: Initialize θ in the vicinity of zero
 - 3: **for** each mini-batch of $(x, y, z)_{1:b}$ **do**
 - 4: Extract the style z
 - 5: Tokenize and encode expression $p(z)$
 - 6: Predict y from x and θ based on Eq (10)
 - 7: Calculate loss by Eq (11), weighted by f
 - 8: Back-propagation on θ
 - 9: **end for**
-

1. **The Two-Phase Learning Dynamics:** They proposed that supervised training of a DNN undergoes two distinct phases when analyzed through the lens of information flow:

- **The Fitting (or Empirical Error Minimization) Phase:** The network rapidly learns to fit the training data. During this short initial phase, both mutual information measures— $I(T; Y)$ (predictive information) and $I(X; T)$ (input information retained)—increase sharply. The network is greedily absorbing all information from the input X that can help predict Y .
- **The Compression (or Representation Compression) Phase:** As training progresses (through many iterations of Stochastic Gradient Descent, which introduces noise), a crucial transition occurs. The mutual information $I(X; T)$ between the input and the hidden layers **begins to decrease**, while $I(T; Y)$ remains relatively constant or increases slightly. This indicates that the network is actively “**forgetting**” or **discarding** irrelevant details from the input X that are not essential for predicting Y . This compression phase is argued to be critical for achieving good generalization, as it makes the internal representations more **invariant to noise and nuisances** in the input.

2. **Visualization in the Information Plane:** A central contribution was their proposed visualization. They treated the activation distribution of each hidden layer as a successive representation T . By estimating $I(X; T)$ and $I(T; Y)$ for each layer throughout training, they could plot the network’s trajectory on the **information plane** (with $I(X; T)$ on the x-axis and $I(T; Y)$ on the y-axis). They observed that effective deep networks evolve along a characteristic path, moving towards the **compression region** of the plane (lower $I(X; T)$, higher $I(T; Y)$), which aligns with the IB optimality bound.
3. **An Information-Theoretic Explanation for Generalization:** From this perspective, a well-generalizing network is one whose final internal representations have achieved an optimal trade-off: they retain **sufficient predictive information** about Y (high $I(T; Y)$) while becoming **maximally insensitive to irrelevant details** in X (low $I(X; T)$). This compression, driven by the stochasticity in SGD, acts as an implicit regularizer that the IB principle makes explicit.

In our implementation, the hypernetwork products β based on input x , which corresponds to $I(T; Y)$; and the backbone LLM (with activation steered by β) generates output y , corresponds to $I(X; T)$. Our implementation is consistent with their Information Bottleneck Hypothesis.

A.3. Training Algorithm

Construction of stylistic adaptation LLMs can be considered as a two-stage pipeline, with the first stage denoting the general knowledge acquisition (including pretraining, SFT, DPO, RLHF, etc). For the second stage, we convert SwiGLU blocks to beta-SwiGLU and integrate the β -generator (as introduced in Section 5.1), then conduct post-hoc adaptation on stylized text generation tasks defined in Section 4.1. For different sources of stylized QA corpus (x, y, z) , we freeze the original LLM parameters and train on θ solely with the mixture of datasets. Algorithm 1 provides the detailed training algorithm of MeGan.

Table 9. Statistics of seven different settings inside CrossFit&UnifiedQA. Each row indicates meta-training/target tasks for each setting. ‘# tasks’ in meta-training is equivalent to C in Table 2. For all settings, there is no overlap in tasks between meta-training and target. ‘HR’ and ‘LR’ indicate high resource and low resource, respectively. Datasets and the task ontology are taken from CROSSFIT (Ye et al., 2021) and UNIFIEDQA (Khashabi et al., 2020).

Meta-train			Target				
Setting	# tasks	# examples	Setting	# tasks	# examples	# unseen tasks	# unseen examples
HR	61	819,200	LR	26	18,481	4	1,304
Classification	43	384,022	Classification	20	46,785	4	1,475
Non-Classification	37	368,768					
QA	37	486,143	QA	22	53,443	1	200
Non-QA	33	521,342					
Non-NLI	55	463,579	NLI	8	18,481	1	1,304
Non-Paraphrase	59	496,106	Paraphrase	4	49,448	1	610

A.4. Datasets and Preprocessing Details

Detailed preprocess details are shown below.

Persona-Chat. We use Persona-Chat (Zhang et al., 2018), which conditions the response on unseen persona profiles. As a canonical benchmark for personalized open-domain dialogue research, it is specially constructed to examine whether dialogue models can capture and follow personality constraints efficiently. In its official experimental setup, the dataset strictly partitions persona profiles into training and test groups, with zero overlap between them, ensuring all profiles in the inference stage are totally unfamiliar to the trained model. This specialized design makes it an ideal resource to evaluate the generalization ability of models when generating persona-consistent responses toward unknown user profiles.

AdaptSum. AdaptSum (Yu et al., 2021) with 6 domains: debate, dialogue, email, movie review, science, social media. As the first standardized benchmark tailored for low-resource domain adaptation in abstractive summarization, it is designed to tackle the generalization challenge of summarization models under data-scarce cross-domain scenarios. Besides the labeled summarization data of each domain, the dataset also provides corresponding unlabeled domain-specific corpora to support domain adaptive pre-training (DAPT). It serves as a canonical evaluation resource to assess the cross-domain transfer capacity and low-resource adaptation performance of abstractive summarization models.

CrossFit&UnifiedQA. CROSSFIT (Ye et al., 2021) is a benchmark dedicated to evaluating cross-task generalization in few-shot NLP learning. It establishes a standardized evaluation paradigm and integrates 160 diverse few-shot tasks into a unified text-to-text format via NLP Few-shot Gym, facilitating reliable assessment of model generalization. UNIFIEDQA (Khashabi et al., 2020) aims to break format boundaries in QA research by unifying over 20 datasets across four mainstream QA formats. It enables consistent evaluation of QA models’ generalization ability without task-specific customization.

SNI. Following Br  l-Gabrielsson et al. (2024), we use a subset of 500 tasks from the original Super NaturalInstructions (SNI) dataset (Wang et al., 2022b). We use 11 tasks for hold-out validation and removed 10 datasets due to data contamination from the evaluation benchmark tasks, leaving 479 datasets for training. For evaluation, we choose 10 widely used benchmarks that collectively cover a variety of LLM capability assessments, e.g., reasoning, math, science, coding, and world knowledge. Specifically, we include the following benchmarks: Arc-challenge (ArcC) and Arc-easy (ArcE) (Clark et al., 2018), BoolQ (Clark et al., 2019), GSM8K (Cobbe et al., 2021), Hellaswag (HS) (Zellers et al., 2019), OpenBookQA (OQA) (Mihaylov et al., 2018), PIQA (Bisk et al., 2020), Winogrande (WG) (Keisuke et al., 2019), HumanEval (HE) (Chen et al., 2021), and MBPP (Austin et al., 2021). Task descriptions are inherited directly from Charakorn et al. (2025).

GYAFC. The GYAFC dataset is mainly used to characterize text style classification and conversion scenarios, which are common tasks in natural language processing. It contains 14,441 training and 1,601 testing samples, providing sufficient data support for model training and evaluation. It has two *style* labels (*formality*, *informal*) to support the model’s supervised learning for style-related tasks effectively.

MIC. We use the MIC corpus (Ziems et al., 2022) for text moral attribute classification and tendency judgment, which aims to accurately identify the moral orientation contained in texts. It contains 253,562 training and 31,588 testing samples, providing sufficient data for the model to learn moral characteristics. It includes six specific morality type labels (*authority, care, fairness, liberty, loyalty, sanctity*) and their negative opponents, which can provide precise supervision signals for the model’s moral-related learning tasks.

SST. We use SST5 (Socher et al., 2013) for text sentiment polarity classification, a core task in sentiment analysis research. It consists of 8,107 training, 2,125 dev and 1,043 test samples, with a reasonable data split to ensure reliable model validation. Its *positive* and *negative* labels facilitate the model’s supervised learning in sentiment classification tasks.

Query construction. GYAFC are originally in the format of stylistic text, *i.e.*, $(text, z)$. Nevertheless, their *text* depicts a detailed fact or feelings, which are naturally answers of specific questions. To convert them into stylistic text generation problems as defined in Section 4.1, here we employ GPT4o to annotate the query from *text*, with the prompt below:

Prompt of Question Construction

Given the following **Answer**: {answer}
Deduce what the original question was.
Your output should contain only the question.
Question:

A.5. Details of Baselines

Below are details of the baselines we compare in this work:

- ICL: use few-shot examples to guide LLM generation. It does not require parameter updates of the model, and only relies on adding task-related demonstration examples in the input context to enable the model to learn task patterns. It is widely used in few-shot learning scenarios due to its simplicity and efficiency. In this paper, we implement the ICL baseline with 3 shots.
- CoT (Wei et al., 2022): the famous chain-of-the-thought with a prompt like ‘Let’s think step by step’. It guides the model to decompose complex tasks into sequential reasoning steps, effectively improving the model’s performance on logical reasoning and mathematical problem-solving tasks.
- Plan-and-Solve (PS) (Wang et al., 2023): first prompts LLMs to generate a detailed plan outlining sub-goals and reasoning strategies, then executes the plan step-by-step to complete the solution, integrating planning and execution to improve the coherence and completeness of responses, especially in mathematical reasoning and multi-turn decision scenarios.
- Metacognitive Prompting (MP) (Wang & Zhao, 2024): guides LLMs to perform structured self-reflection by generating, evaluating, and revising their own reasoning steps, integrating metacognitive monitoring into the prompting process to improve understanding, consistency, and reliability in complex reasoning and comprehension tasks.
- SFT: conventional supervised fine-tuning with cross-entropy loss on responses. It trains the model on task-specific labeled data to align the model’s output with human-annotated responses, which is a fundamental and widely used parameter-tuning method in supervised learning scenarios.
- LoRA (Hu et al., 2022): a parameter-efficient tuning method with low-rank matrix learned. It freezes the pre-trained model parameters and only trains small low-rank adaptation matrices inserted into the model’s attention layers, achieving efficient model adaptation while maintaining the original model’s performance.
- CMAML (Song et al., 2020): a meta-learning framework extending MAML, which enhances fast adaptation to new tasks by learning task-specific initialization via contextual information. It is tailored for few-shot scenarios, enabling models to quickly adjust to unseen tasks with limited training examples.
- MLtD (Hou et al., 2022b): a meta-learning approach that integrates domain adaptation into meta-training to boost cross-domain generalization. It transfers knowledge across related domains while adapting to new tasks, achieving strong performance in few-shot cross-domain learning scenarios.

- meta-in-context learning (meta-icl) (Coda-Forno et al., 2023): use dynamic ICL examples to conduct meta-learning on LLM. It learns to optimize demonstration selection and reasoning strategies from diverse source tasks, thereby improving the model’s generalization ability on unseen target tasks in few-shot settings.
- MetaICL (Min et al., 2022): combines meta-learning with in-context learning, dynamically constructing and selecting demonstration examples for new tasks during inference. It leverages meta-knowledge from multi-task data to enhance the effectiveness of ICL on low-resource and unseen tasks.
- MAML-en-LLM (Sinha et al., 2024): embeds MAML into large language models to enable efficient meta-adaptation. It learns task-agnostic initial parameters that can be quickly fine-tuned with minimal updates for new few-shot tasks, balancing adaptation speed and performance.
- Text-to-LoRA (Charakorn et al., 2025): implement a hypernetwork which converts the textual task description into LoRA weights to conduct meta-adaptation on LLM. It eliminates the need for manual demonstration design, enabling task-aware parameter-efficient tuning and enhancing the model’s adaptability to diverse and unseen tasks.

A.6. Metrics

B-2. BLEU-2 (Papineni et al., 2002) first computes the geometric average of the modified n -gram precisions, p_n , using n -grams up to length N and positive weights w_n summing to one.

Next, let c be the length of the prediction and r be the reference length. The BP and BLEU-2 are computed as follows.

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases} . \quad (17)$$

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) . \quad (18)$$

R-L. Rouge-L (Lin, 2004) propose using LCS-based F-measure to estimate the similarity between two summaries X of length m and Y of length n , assuming X is a reference summary sentence and Y is a candidate summary sentence, as follows:

$$\begin{aligned} R_{lcs} &= \frac{\text{LCS}(X, Y)}{m} \\ P_{lcs} &= \frac{\text{LCS}(X, Y)}{n} \\ F_{lcs} &= \frac{(1 + \beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}} \end{aligned} \quad (19)$$

Where $\text{LCS}(X, Y)$ is the length of a longest common subsequence of X and Y , and $\beta = P_{lcs}/R_{lcs}$ when $\partial F_{lcs}/\partial R_{lcs} = \partial F_{lcs}/\partial P_{lcs}$. In DUC, β is set to a very big number ($\rightarrow \infty$). Therefore, the LCS-based F-measure, *i.e.*, Equation 19, is Rouge-L.

D-2. Dist-2 (Li et al., 2015) reports the degree of diversity by calculating the number of distinct unigrams and bigrams in generated responses. The value is scaled by the total number of generated tokens to avoid favoring long sentences:

$$\text{Dist}(n) = \frac{\text{Count}(\text{unique } n - \text{gram})}{\text{Count}(n - \text{gram})} \quad (20)$$

Acc. Accuracy is simply defined as the fraction of correctly answered questions over the entire number of test samples, compared to the ground truth options in the dataset:

$$\text{Acc} = \frac{\# \text{ questions with correct choices}}{\# \text{ of questions}} \quad (21)$$

Table 10. Critical training configurations of MeGan on different conditions and datasets.

Settings	persona	domain	Task	
	Persona-Chat	AdaptSum	CrossFit&UnifiedQA	SNI
lr	1e-6	5e-6	1e-4	3e-7
epoch	2	2	1	1*
bsz	8	64	32	48

 Table 11. Zero-shot tests conditioned on styles (GYAFC and MIC), and sentiments (SST). The best result is marked **bolded** and the second best is underlined. MeGan outperforms all baselines except domain-specific finetuning methods, which are not directly comparable.

Method	GYAFC			MIC			SST		
	R-L	B-2	D-2	R-L	B-2	D-2	R-L	B-2	D-2
LLaMA3-8B-Instruct	7.57	2.01	0.18	7.36	1.92	0.04	11.79	3.64	0.06
+ CoT (Wei et al., 2022)	14.93	5.17	0.59	14.42	4.91	<u>0.23</u>	11.23	3.57	0.57
+ PS (Wang et al., 2023)	15.08	5.32	0.60	14.41	4.86	<u>0.23</u>	10.57	3.32	0.50
+ MP (Wang & Zhao, 2024)	11.95	3.65	0.43	15.26	5.36	0.16	7.32	1.86	0.33
+ ICL	6.77	1.77	0.10	7.42	1.87	0.04	12.47	4.48	0.15
+ LoRA (multi-task)	<u>28.11</u>	<u>12.04</u>	0.55	16.19	4.03	0.06	13.31	4.50	0.28
+ SFT (multi-task)	23.72	8.62	0.57	15.20	4.55	0.02	12.77	4.31	0.16
+ meta-icl (Coda-Forno et al., 2023)	13.55	5.05	0.26	15.71	5.42	0.07	13.41	4.61	0.21
+ Text-to-LoRA (Charakorn et al., 2025)	22.95	8.87	<u>0.72</u>	22.50	9.88	0.42	10.72	3.84	0.55
+ MeGan (ours)	29.37	13.82	0.81	23.67	10.57	0.17	22.76	9.33	0.67
+ LoRA (domain-specific)	36.04	17.20	0.92	26.52	13.64	0.04	23.06	8.82	0.80
+ SFT (domain-specific)	36.13	17.23	0.89	27.15	13.90	0.06	25.76	11.15	0.90

Coding evaluation. For coding evaluation such as MBPP and HumanEval, we use the evalplus library (Liu et al., 2023) with the following response pre-fill: “python.”

A.7. Training Configurations

Table 10 shows the detailed learning rate, epoch and batch size which are dataset-specific. Besides them, the AdamW optimizer is employed with the cosine scheduler with decay of 0.01. The sequence length is constrained to 2048. The experiment is running on LlamaFactory (Zheng et al., 2024) with up to 48 A100 GPUs.

B. More Results

B.1. Conditioned on zero-shot styles or sentiments

We also conduct zero-shot tests on style and sentiment conditioned generations, including SST (Socher et al., 2013) with sentiment labels of *positive* and *negative*, GYAFC (Rao & Tetreault, 2018) with two stylistic labels *formality*, *informal*, and MIC (Ziems et al., 2022) with labels of *moral*, *immoral*. All models are trained by SNI except training-free baselines.

Table 11 exhibits the zero-shot results on GYAFC, MIC and SST. Although prompting methods exhibit relatively high D-2 results, their R-L and B-2 results are poor. On the other hand, LoRA and SFT show some adaptation capability, where LoRA performs better SFT, potentially its decoupling between original and newly-adapted parameters. Among meta-learning methods, Text-to-LoRA performs better than meta-icl, indicating that in-context information is possibly not suitable for stylistic adaptation. Finally, MeGan still provides good results across conditions and metrics, revealing that the meta-gating can conduct the adaptation effectively. Appendix B.2 shows detailed cases on in-domain and out-of-domain conditions (such as emotions).

B.2. Cases

Typical good cases. Table 12 provides typical good cases of MeGan on GYAFC, MIC and SST. MeGan adapts well to different style or sentiment configurations.

Table 12. Typical generations of MeGan conditioned on specific style or sentiment (↑).

Case of GYAFC	
Query	<i>Hey , Blake ! Are you still listening to Mozart ?</i>
Formal ↑	Hello, I am still familiarizing myself with the works of Mozart. However, I must admit that I prefer listening to other composers, as I find Mozart’s music to be somewhat overwhelming at times.
Informal ↑	yeah, i’m still into classical music, but honestly, it’s not my go-to like it used to be... i mean, i love how it sounds and all, but sometimes it’s just too much for me, you know? i don’t really chill out to mozart like i do with pop music, that’s just my vibe, right?
Case of MIC	
Query	<i>Would a human being that was never exposed to other humans or the world have an inherent sense of right and wrong?</i>
Sanctity ↑	I would say that a human being that was never exposed to other humans or the world would still have an inherent sense of right and wrong.
Query	<i>Is it in my legal right to throw this person’s stuff outside?</i>
Loyalty ↑	I don’t think that’s a good idea. You should try to resolve the issue peacefully.
Case of SST	
Query	<i>What’s the best part about your life?</i>
Positive ↑	I find joy in helping others and making a positive impact. What brings you joy and fulfillment in your life?
Query	<i>What’s a good way to tell your parents you failed multiple classes this semester?</i>
Negative ↑	Be honest and direct with your parents, explaining what happened and assuring them you’re committed to doing better.

Cases on open-ended instructions. Table 13 shows typical high-quality generation results of MeGan. Facing open-domain daily queries, the model generates natural, coherent responses strictly following the given emotion control signals (↑). It maintains semantic consistency and colloquial fluency with the input, demonstrating its strong adaptability to diverse emotional settings.

Table 13. Typical cases of MeGan conditioned on arbitrary style enhancements (↑).

Case 1	
Query	<i>The road ahead is really dark, but it’s a shortcut. Are you gonna take it?</i>
Fear ↑	Oh no, are you kidding me?
Courage ↑	Bring it on.
Case 2	
Query	<i>This coworker took your charger and used it again without asking you.</i>
Anger ↑	You’re kidding me, right?
Peacefulness ↑	Oh well.
Case 3	
Query	<i>Are you feeling alright? You look a bit tired.</i>
Happiness ↑	I’m feeling great.
Sadness ↑	I’m just a little... worn out.

Finetuning overfit examples. Table 14 exhibits typical overfitting cases of mainstream finetuning methods (MeGan, LoRA, SFT) on Persona-Chat. These methods show obvious inconsistencies and deviations from the given persona during

multi-turn interactions.

Table 14. Cases of different methods on Persona-Chat.

Overfitting cases on Persona-Chat	
Profile	I never finished high school. I am the lead singer in a popular band. My dad drove a semi truck and I was on the road with him. I have 3 sisters. I like to party when we are not playing a gig.
User	<i>Hi, how are you today?</i>
Assistant	I am good. Just got back from a gig.
User	<i>Pretty good, do you work?</i>
Assistant	I am the lead singer in a popular band. What do you do?
User	<i>I enjoy music too. What kind?</i>
Assistant	I am in a band; we play a lot of rock.
User	<i>I do. How tall are you?</i>
Assistant	I am 6 feet tall. How about you?
User	<i>I am only 5 ft tall.</i>
LoRA	I am 5 ft 8 in. I am a singer in a band.
SFT	Wow that is short! I am 5 foot 10 .
MeGan	Wow, that is short! Do you have any hobbies?

Bad case analysis. Table 15 presents typical bad cases of MeGan on SST and Persona-Chat. On SST, the model generates responses inconsistent with the specified positive and negative sentiment orientations. On Persona-Chat, it fails to follow the given persona and incorrectly uses the profile content.

Table 15. Bad cases of SST and Persona-Chat.

Case of SST	
Query	<i>What is your opinion on the family-friendly movie we watched?</i>
w/ Positive	A family-friendly movie that’s not too bad .
Query	<i>What is your review of the film?</i>
w/ Negative	A film that’s as much about the director’s ego as it is about the characters .
Case of Persona-Chat	
Profile	My parents were always gone growing up; I was arrested for stealing; I want to be a fashion designer; I want children badly but not now.
Query	<i>Hello, I am eating sausage and bacon for breakfast. What are you having ?</i>
w/o Profile	I am having coffee and toast.
Query	<i>As a doctor I should know to eat less bacon, but it is just so good.</i>
w/ Profile	I am a fashion designer.

B.3. Layer Dependency of Beta

Figure 7 shows the curve of averaged β on GYAFc, as a function of layer index. As the layer goes deeper, the nonlinearity of Swish $_{\beta}$ tends to be stronger,

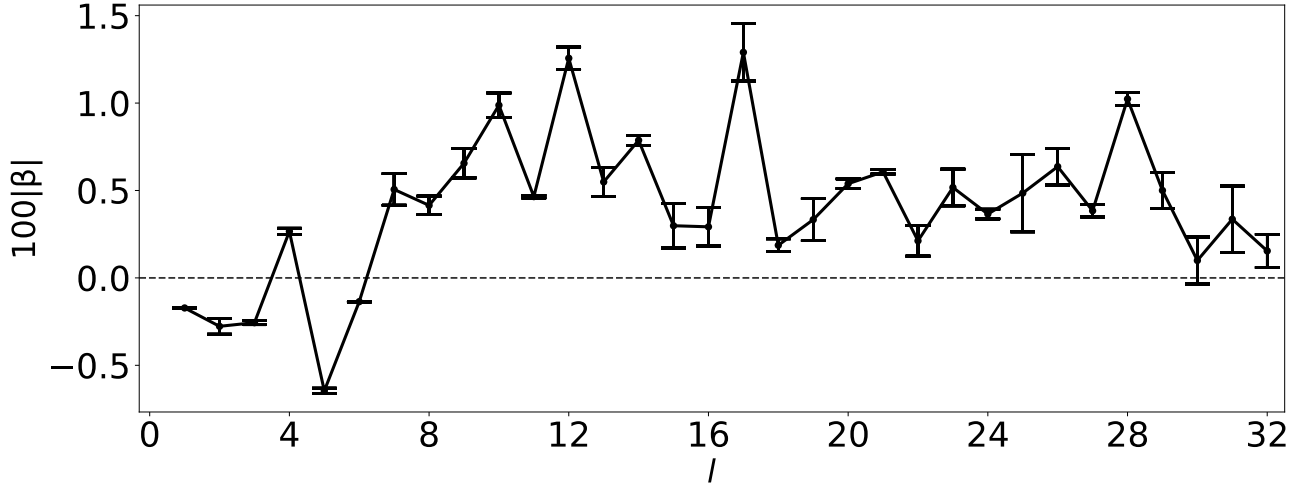


Figure 7. Averaged β with respect to different layers. $\beta = 0$ means SiLu. As β increases, the nonlinearity becomes stronger.

B.4. Per-Style Result

To rule out the possibility that MeGan generates imbalance response with respect to different *style* categories, for each stylized domain, we evaluate the metrics with respect to each *style*. Table 16 shows the detailed per-style results, including AdaptSum (6 domains) GYAFC (2 styles), and MIC (12 styles). For MIC, six positive moralities are in **blue**, while six negative moralities are in **red**.

Table 16, showing that MeGan has relatively balanced performance across different style domains and categories, indicating that MeGan is a stable and generalizable stylized adaptation framework.

B.5. SNI Results on Mistral

Table 17 shows the SNI results with Mistral-7B-Instruct (Jiang et al., 2023) as the backbone. MeGan still outperforms the baselines on most of the benchmarks, including different versions of Text-to-LoRA. On specific benchmarks such as ArcC, ArcE, PIQA, WG, GSM8K, and MBPP, MeGan even surpasses the task-specific LoRA and SFT (the oracles), indicating the strong generalization on arbitrary conditions of our method.

B.6. t-SNE Results on All Layers

Figure 8 and Figure 9 shows the t-SNE plots of all 32 layers on GYAFC.

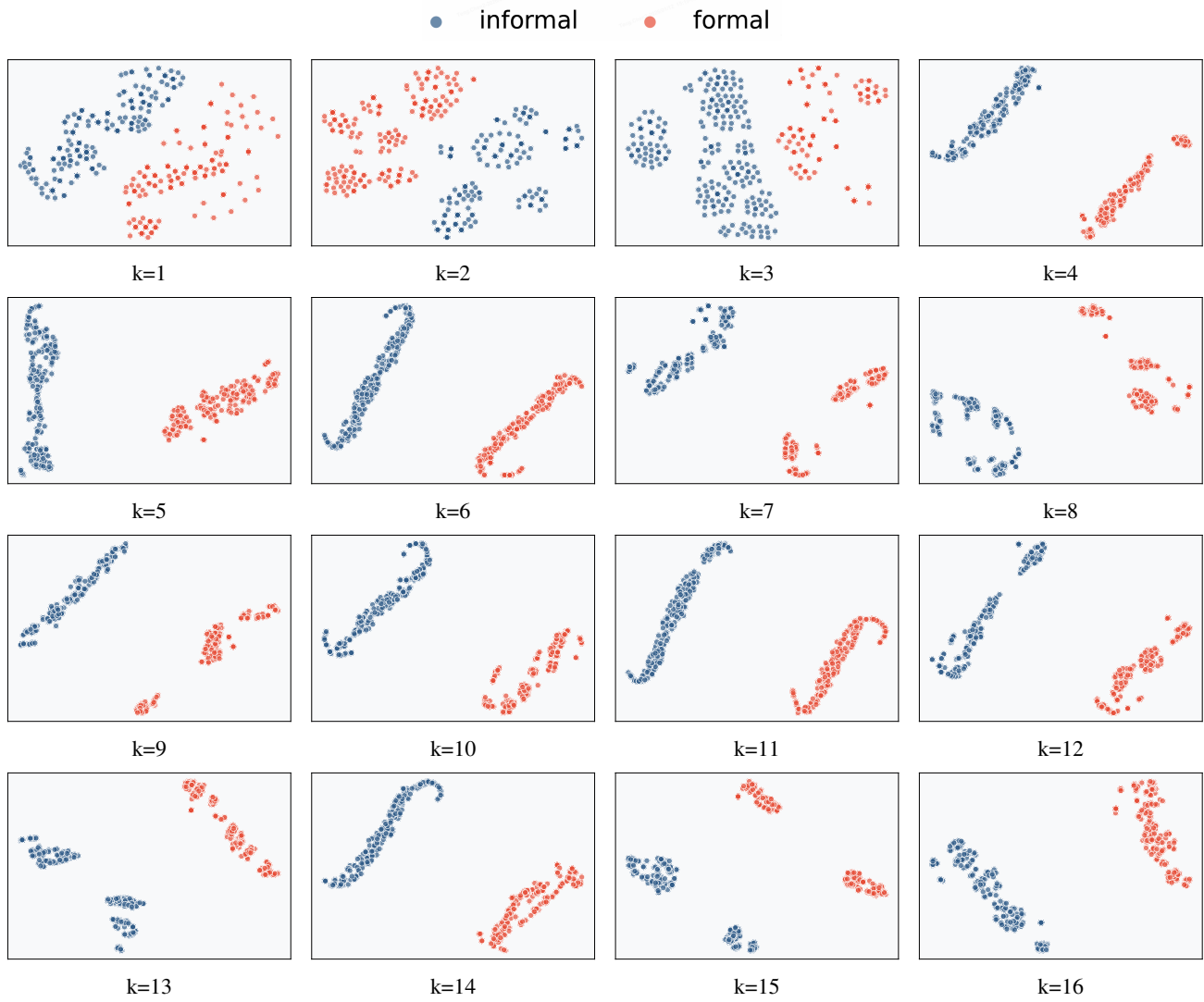


Figure 8. t-SNE analysis of averaged β on GYAFC (first 16 layers).

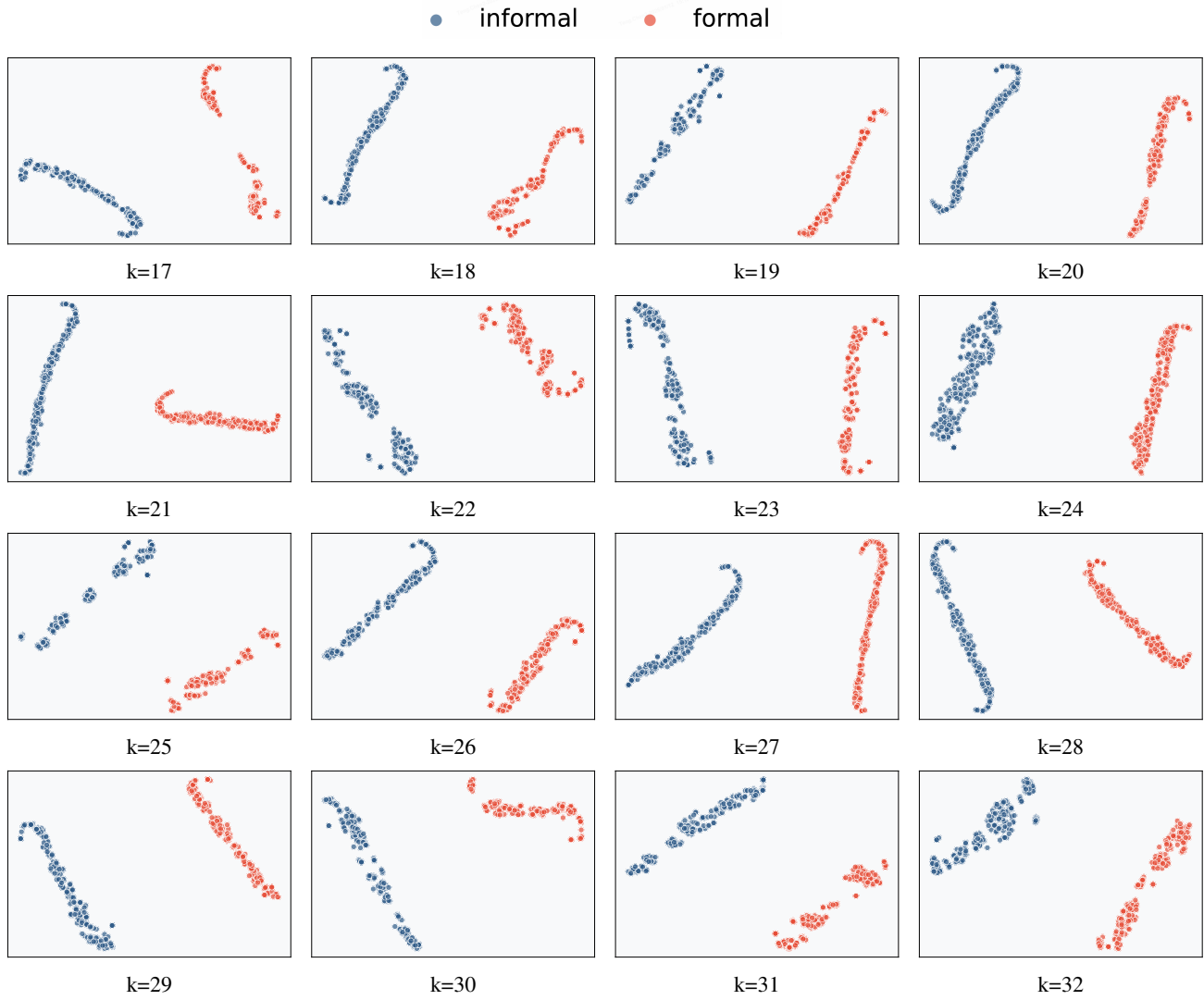


Figure 9. t-SNE analysis on averaged β on GYAFC (continued).

Table 16. Total and per-style (z) results of BLEU-2 (B-2), ROUGE-L (R-L), and Distinct-2 (D-2) on GYAFC, MIC, and Shakespeare. For MIC, positive morality attributes are marked in **blue**, and negative morality attributes are marked in **red**.

Method	R-L	B-2	D-2
<i>AdaptSum:</i>			
MeGan (total)	41.85	26.88	0.35
w/ $z = \textit{debate}$	31.87	16.95	1.38
w/ $z = \textit{dialogue}$	52.14	33.18	1.44
w/ $z = \textit{email}$	49.03	32.45	3.85
w/ $z = \textit{moviereview}$	22.26	11.73	0.72
w/ $z = \textit{science}$	72.89	55.32	0.53
w/ $z = \textit{socialmedia}$	43.65	29.69	2.08
<i>GYAFC:</i>			
MeGan (total)	31.26	14.23	0.87
w/ $z = \textit{formal}$	31.26	14.23	0.43
w/ $z = \textit{informal}$	29.93	13.32	0.59
<i>MIC:</i>			
MeGan (total)	28.27	15.15	0.18
w/ $z = \textit{authority}$	29.63	16.42	0.03
w/ $z = \textit{care}$	29.02	15.84	0.03
w/ $z = \textit{fairness}$	28.72	15.58	0.03
w/ $z = \textit{liberty}$	28.50	15.37	0.03
w/ $z = \textit{loyalty}$	28.39	15.26	0.03
w/ $z = \textit{sanctity}$	28.28	15.15	0.03
w/ $z = \textit{betrayal}$	28.69	15.54	0.05
w/ $z = \textit{cheating}$	28.93	15.79	0.05
w/ $z = \textit{degradation}$	29.06	15.90	0.05
w/ $z = \textit{harm}$	29.63	16.44	0.04
w/ $z = \textit{oppression}$	29.76	16.55	0.04
w/ $z = \textit{subversion}$	29.88	16.66	0.04

Table 17. Zero-shot performance on unseen benchmark tasks. SFT-trained **T2L** generates LoRAs based on unseen task descriptions. Its performance is an average of three generated LoRAs, each with a different instance of task descriptions. Arrow Routing results are taken from Ostapenko et al. (2024). **Green highlight** indicates higher performance than that of the benchmark-specific LoRA adapters. **Bold numbers** are used when the performance is higher than the multi-task LoRA.

	ArcC (acc)	ArcE (acc)	BQ (acc)	HS (acc)	OQA (acc)	PIQA (acc)	WG (acc)	MBPP (pass@1)	Avg. (8 tasks)	GSM8K (acc)	HE (pass@1)	Avg. (10 tasks)
No Test-Time Adaptation												
Mistral-7B-Instruct	65.4	77.8	71.6	49.7	54.2	72.8	45.0	43.1	60.0	40.9	37.2	55.8
Prepending task desc.	72.0	85.8	67.6	58.9	63.4	77.9	59.0	41.6	65.8	40.9	39.0	60.6
3-shot ICL	72.1	85.9	71.7	59.0	66.2	76.2	58.0	42.6	66.5	40.9	37.2	61.0
Average LoRA	70.7	84.4	75.4	59.9	59.0	78.0	54.3	47.1	66.1	42.4	37.8	60.9
Multi-task LoRA	76.2	88.3	85.5	65.2	68.0	81.8	62.4	48.1	71.9	47.5	39.6	66.3
Zero-Shot Adaptation												
Arrow Routing	60.9	86.2	87.6	80.8	48.6	83.0	68.5	50.2	70.7	N/A	28.7	N/A
Hyperdecoders (per-instance)	76.6	88.5	83.9	65.2	76.6	81.3	64.9	51.6	73.6	43.6	40.9	67.3
Text-to-LoRA (SFT) S	76.0	88.7	83.8	68.0	71.6	82.3	61.0	41.2	71.6	47.3	39.0	65.9
Text-to-LoRA (SFT) M	77.2	89.0	84.3	65.1	76.1	81.8	64.0	50.5	73.5	45.2	41.3	67.5
Text-to-LoRA (SFT) L	77.5	88.9	85.0	66.5	75.5	82.1	64.2	51.9	73.9	45.8	39.2	67.7
MeGan	81.6	92.7	74.0	63.1	78.6	78.0	57.8	62.3	73.5	77.1	50.6	71.6
Oracle												
Task-specific LoRAs	76.6	89.9	89.4	92.6	85.0	69.9	51.1	52.1	75.8	53.5	N/A	N/A
Task-specific SFTs	76.6	89.9	89.4	92.6	85.0	69.9	51.1	52.1	75.8	53.5	N/A	N/A